Glossary

Most of the terms in this glossary appear at several or many places in the text of this book. Many terms that are used broadly in biology or are used in this book only near their definition in the text are not included here.

A

absolute fitness *See* **relative fitness**.

active trend *See* **driven trend**.

adaptation A process of genetic change in a population whereby, as a result of natural selection, the average state of a character becomes improved with reference to a specific function, or whereby a population is thought to have become better suited to some feature of its environment. Also, *an* adaptation: a feature that has become prevalent in a population because of a selective advantage conveyed by that feature in the improvement in some function.

adaptive landscape A metaphor for the relationship, or mathematical function, between mean fitness of a population and the allele frequencies at one or more loci that affect fitness. Possible populations with allele frequencies that maximize mean fitness are represented as peaks on the metaphorical landscape.

adaptive peak That allele frequency, or combination of allele frequencies at two or more loci, at which the mean fitness of a population has a (local) maximum. Also, the mean phenotype (for one or more characters) that maximizes mean fitness. An **adaptive valley** is a set of allele frequencies at which mean fitness has a minimum.

adaptive radiation Evolutionary divergence of members of a single phylogenetic lineage into a variety of different adaptive forms; usually the taxa differ in the use of resources or habitats, and have diverged over a relatively short interval of geological time. The term **evolutionary radiation** describes a pattern of rapid diversification without assuming that the differences are adaptive.

adaptive valley *See* **adaptive peak**.

adaptive zone A set of similar **ecological niches** occupied by a group of (usually) related species, often constituting a higher taxon.

additive effect The magnitude of the effect of an allele on a character, measured as half the phenotypic difference between homozygotes for that allele compared with homozygotes for a different allele.

additive genetic variance That component of the **genetic variance** in a character that is attributable to additive effects of alleles.

allele One of several forms of the same gene, presumably differing by mutation of the DNA sequence. Alleles are usually recognized by their phenotypic effects; DNA sequence variants, which may differ at several or many sites, are usually called **haplotypes**.

allele frequency The proportion of gene copies in a population that are a given allele; i.e., the probability of finding this allele when a gene is taken randomly from the population; also called **gene frequency**.

allometric growth Growth of a feature during ontogeny at a rate different from that of another feature with which it is compared.

allopatric Of a population or species, occupying a geographic region different and separated from that of another population or species. *Cf.* **parapatric**, **sympatric**.

allopatric speciation Speciation by genetic divergence of allopatric populations of an ancestral species; contrasted with **parapatric** and **sympatric speciation**, in which divergence occurs in parapatry or sympatry (q.v.).

allopolyploid A **polyploid** in which the several chromosome sets are derived from more than one species.

allozyme One of several forms of an enzyme encoded by different alleles at a locus, that are distinguished by gel electrophoresis.

alternative mating strategies Different mating behaviors and morphologies that are maintained as a stable polymorphism by negative frequency-dependent selection.

alternative splicing Splicing of different sets of exons from RNA transcripts to form mature transcripts that are translated into different proteins (thus allowing the same gene to encode different proteins).

altruism Conferral of a benefit on other individuals at an apparent cost to the donor.

anagenesis Evolutionary change of a feature within a lineage over an arbitrary period of time.

ancestral character state An evolutionarily older character state, relative to another (derived) state that has evolved from it in one or more lineages.

aneuploid Of a cell or organism, possessing too many or too few homologous chromosomes, relative to the normal (**euploid**) set.

anisogamy The condition of having two types of gametes of different sizes and forms, one large and immobile (the egg), the other small and usually mobile (the sperm).

antagonistic pleiotropy Contrasting effects of a gene on two different characters, such that the effect of an allele substitution on one character increases fitness, but the effect on the other character decreases fitness.

antagonistic selection A source of natural selection that opposes another source of selection on a trait.

apomixis Parthenogenetic reproduction in which an individual develops from one or more mitotically produced cells that have not experienced recombination or syngamy.

apomorphic Having a **derived** character or state, with reference to another character or state. *See* **synapomorphy**.

aposematic Coloration or other features that advertise noxious properties; warning coloration.

artificial selection Selection by humans of a deliberately chosen trait or combination of traits in a (usually captive) population; differing from natural selection in that the criterion for survival and reproduction is the trait chosen, rather than fitness as determined by the entire genotype.

asexual Pertaining to reproduction that does not entail meiosis and syngamy.

assortative mating Nonrandom mating on the basis of phenotype; usually refers to positive assortative mating, the propensity to mate with others of like phenotype.

autopolyploid A **polyploid** in which the several chromosome sets are derived from the same species.

autosome A chromosome other than a sex chromosome.

B

back mutation Mutation of an allele back to the allele from which it arose by an earlier mutation.

background extinction A long-prevailing rate at which taxa become extinct, in contrast to the highly elevated rates that characterize mass extinction.

background selection Elimination of deleterious mutations in a region of the genome; may explain low levels of neutral sequence variation.

balancing selection A form of natural selection that maintains polymorphism at a locus within a population.

base pair substitution As usually used in this book, a base pair that, because of genetic drift or natural selection, has replaced another base pair at a specific DNA site in a population or species.

behavioral ecology The study of the evolution of behaviors, often in relation to the environment, including other members of the same species.

behavioral isolation *See* **sexual isolation**.

benthic Inhabiting the bottom, or substrate, of a body of water. *Cf.* **planktonic**.

biogeographic realm Major geographic regions of Earth that have characteristic animal and plant taxa.

biogeography The study of the geographic distribution of organisms.

biological homology Commonality of different traits, among or within species, based on a shared genetic basis and developmental pathway; the traits are often, but not always, homologous in the usual phylogenetic sense. *See* **homology**.

biological species A population or group of populations within which genes are actually or potentially exchanged by interbreeding, and which are reproductively isolated from other such groups.

bottleneck A severe, temporary reduction in population size.

breeder's equation The equation that predicts that the evolutionary change in the mean of a quantitative trait resulting from selection in one generation is equal to the product of the trait's **heritability** and the **selection differential**.

C

C-value paradox The lack of correlation between the DNA content of eukaryotic genomes and a given organism's phenotypic complexity (i.e., the genome of a less complex eukaryotic organism, such as a plant, may contain far more DNA than that of a more complex organism, such as a human). The paradox is explained by the amount of noncoding DNA sequences in a genome.

Cambrian explosion The first appearance in the fossil record of many animal phyla, within a relatively short (<20 million years) interval.

canalization The evolution of internal factors during development that reduce the effect of perturbing environmental and genetic influences, thereby constraining variation and consistently producing a particular (usually wild-type) phenotype.

candidate gene A gene postulated to be involved in the evolution of a particular trait based on its mutant phenotype or the function of the protein it encodes.

carrying capacity The population density that can be sustained by limiting resources.

category In taxonomy, one of the ranks of classification (e.g., genus, family). *Cf.* **taxon**.

cDNA (complementary DNA) A DNA copy of an mRNA made using reverse transcriptase isolated from a retrovirus.

cDNA library A collection of cDNAs, representing the transcriptome (all of the mRNAs expressed) of a tissue or whole organism at a particular life history stage, created by isolating cDNA, cloning it into circular DNA plasmids and propagating it in bacterial cells.

character A feature, or trait. *Cf.* **character state**.

character displacement As originally used, a pattern of geographic variation in which a character differs more greatly between sympatric than between allopatric populations of two species; now often used for the evolutionary process of accentuation of differences between sympatric populations of two species as a result of the reproductive or ecological interactions between them.

character state One of the variant conditions of a character (e.g., yellow versus brown as the state of the character "color of snail shell").

cheat In **behavioral ecology**, a behavior that allows individuals to receive the fitness benefit of the altruistic behavior of others without paying the fitness cost that the altruistic individuals pay.

chimeric gene A gene that consists of parts of two or more different ancestral genes.

chronospecies A segment of an evolving lineage preserved in the fossil record that differs enough from earlier or later members of the lineage to be given a different binomial (name). Not equivalent to biological species.

***cis*-regulatory element** A noncoding DNA sequence in or near a gene required for proper spatiotemporal expression of that gene, often containing binding sites for transcription factors. *Cf.* **control region**, ***trans*-regulatory element**.

clade The set of species descended from a particular ancestral species.

cladistic Pertaining to branching patterns; a cladistic classification classifies organisms on the basis of the historical sequences by which they have diverged from common ancestors.

cladogenesis Branching of lineages during phylogeny.

cladogram A branching diagram depicting relationships among taxa; i.e., an estimated history of the relative sequence in which they have evolved from common ancestors.

cline A gradual change in an allele frequency or in the mean of a character over a geographic transect.

clonal interference During the increase in frequency of two different beneficial alleles or genotypes, the elimination of one by another that has greater fitness.

clone A lineage of individuals reproduced asexually, by mitotic division.

coadapted gene pool A population or set of populations in which prevalent genotypes are composed of alleles at two or more loci that confer high fitness in combination with each other, but not with alleles that are prevalent in other such populations.

coalescence Derivation of the gene copies in one or more populations from a single ancestral copy, viewed retrospectively (from the present back into the past).

codon A nucleotide triplet that encodes an amino acid or acts as a "stop" signal in translation.

codon bias Nonrandom usage of synonymous codons to encode a given amino acid.

coefficient of relationship The probability that an allele carried by one individual is also carried by a related individual.

coefficient of selection The proportion by which the average fitness of individuals of one genotype differs from that of a reference genotype.

coevolution Strictly, the joint evolution of two (or more) ecologically interacting species, each of which evolves in response to selection imposed by the other. Sometimes used loosely to refer to evolution of one species caused by its interaction with another, or simply to a history of joint divergence of ecologically associated species.

commensalism An ecological relationship between species in which one is benefited but the other is little affected.

common ancestor A lineage (often designated as a taxon) from which two or more descendant lineages evolved.

common garden A place in which (usually conspecific) organisms, perhaps from different geographic populations, are reared together, enabling the investigator to ascribe variation among them to genetic rather than environmental differences. Originally applied to plants, but now more generally used to describe any experiment of this design.

comparative genomics Analysis of similarities and differences between the genomes of different species.

comparative method A procedure for inferring the adaptive function of a character by correlating its states in various taxa with one or more variables, such as ecological factors hypothesized to affect its evolution.

competition An interaction between individuals of the same species or different species whereby resources used by one are made unavailable to others.

competitive exclusion Extinction of a population due to competition with another species.

competitive exclusion principle The theoretical assertion that one of two ecologically identical species will eventually replace the other by competition.

concerted evolution Maintenance of a homogeneous nucleotide sequence among the members of a gene family, which evolves over time.

condition-dependent indicator A characteristic, usually used in behavioral display, that is correlated with, and therefore indicates, the health or physiological vigor ("condition") of an individual.

conflict In **behavioral ecology**, interactions between individuals that increase the fitness of one individual at a cost to the fitness of the other. **Altruistic** and **spiteful** interactions include conflict.

congeneric Belonging to the same genus.

conservative characters Features that evolve slowly and are retained with little or no change for long periods of evolutionary time.

conspecific Belonging to the same species.

constraints Properties of organisms or their environment that tend to retard evolution of a feature or to direct its evolution along some paths rather than others.

control regions Untranscribed regions of the genome to which products of other genes bind, and which determine transcription of specific genes.

convergent evolution (convergence) Evolution of similar features independently in different evolutionary lineages, usually from different antecedent features or by different developmental pathways.

cooperation In **behavioral ecology**, interactions between individuals that are either **mutualistic** or **altruistic**, so that the actor enhances the fitness of the recipient.

co-option The evolution of a function for a gene, tissue, or structure other than the one it was originally adapted for. At the gene level, used interchangeably with **recruitment** and, occasionally, **exaptation**.

Cope's rule A proposed generalization that individual body size in animals tends to increase during evolution.

copy number variants Refers to variation among conspecific individuals in the number of duplicates (copies) of a DNA sequence.

correlated selection Natural selection for specific combinations of traits, such that selection on one trait is correlated with selection on the other.

correlation A statistical relationship that quantifies the degree to which two variables are associated. For *phenotypic correlation*, *genetic correlation*, *environmental correlation* as applied to the relationship between two traits, see Chapter 6.

cost A reduction in fitness caused by a correlated effect of a feature that provides an increment in fitness (i.e., a benefit).

cost of reproduction Reduction of an individual's future fitness (survival and/or future reproduction) caused by reproductive activity.

cost of sex Usually refers to a reduced rate of population growth of a sexual compared to an asexual population, owing to production of males.

creationism The doctrine that each species (or perhaps higher taxon) was created separately, essentially in its present form, by a supernatural Creator.

crown group A taxon, distinguished by derived character states, that has descended from an ancestral group (**stem group**) that may bear a different name.

cultural evolution Changes in the frequency of nongenetic cultural traits within and among populations, based on processes such as nonrandom imitation.

cytoplasmic male sterility (CMS) The sterilization of male function in an otherwise hermaphroditic individual as the result of factors transmitted with the cytoplasm, typically mitochondrial in origin.

D

deleterious mutation A mutation that reduces fitness.

deme A local population; usually, a small, panmictic population.

demographic Pertaining to processes that change the size of a population (i.e., birth, death, dispersal).

de novo genes Coding DNA sequences that originate from noncoding DNA.

density-dependent Affected by population density.

derived character (state) A character (or character state) that has evolved from an antecedent (ancestral) character or state.

deterministic Causing a fixed outcome, given initial conditions. *Cf.* **stochastic**.

developmental arrest A halting of the development of a morphological structure, resulting in a final adult phenotype that lacks the structure or bears an immature form of the structure. This can also refer to developmental arrest at the level of the entire organism, resulting in an adult that resembles the juvenile form of an ancestral or related species (i.e., **paedomorphosis**).

developmental circuit *See* **developmental pathway**.

developmental constraint A restriction that prevents the appearance of certain structures or traits due to the inability of an organism's developmental system to produce them.

developmental pathway A sequence of gene expression through developmental time, involving both gene regulation and the expression of gene products that provide materials for and regulate morphogenesis, resulting in the normal development of a tissue, organ, or other structure. Also called **developmental circuit**.

differential gene expression Differences in the time, location, and/or quantitative level at which a gene expresses the protein it encodes. Differential gene expression involves differences between species, developmental stages, or physiological states in the specific cells, tissues, structures, or body segments that express a given gene.

dioecious Of a species, consisting of distinct female and male individuals.

diploid Of a cell or organism, possessing two chromosome complements. *See also* **haploid**, **polyploid**.

direct benefit (direct fitness benefit) A fitness increment accrued by an individual by performing an action or receiving the action.

direct development A life history in which the intermediate larval stage is omitted and development proceeds directly from an embryonic form to an adult-like form. *Cf.* **indirect development**.

direct fitness *See* **inclusive fitness**.

direct response to selection The component of evolutionary change in the mean of a trait that results from selection acting directly on that trait. *See also* **indirect response to selection**.

direct selection Selection that acts directly on a locus. *See also* **indirect selection**.

directional selection Selection for a value of a character that is higher or lower than its current mean value.

disparity The magnitude of variation in morphological or other phenotypic characters among species in a clade or taxon.

dispersal In population biology, movement of individual organisms to different localities; in biogeography, extension of the geographic range of a species by movement of individuals.

disruptive selection Selection in favor of two or more phenotypes and against those intermediate between them; also called **diversifying selection**.

divergence The evolution of increasing difference between lineages in one or more characters.

diversification An evolutionary increase in the number of species in a clade, usually accompanied by divergence in phenotypic characters.

diversifying selection *See* **disruptive selection**.

diversity-dependent factors Processes that have a stronger effect on per capita rates of speciation or extinction when the diversity of species is greater.

d_N/d_S ratio The ratio of the number of nonsynonymous substitutions per nonsynonymous site (d_N) and the number of synonymous substitutions per synonymous site (d_S). Values of this ratio smaller than one are consistent with **purifying selection**, while values greater than one suggest the action of **positive selection**.

Dobzhansky-Muller (DM) incompatibility Reduction in the fitness of a hybrid because of interaction between certain alleles in one parent population with specific alleles at other loci in the other parent population.

Dollo's law A biological generalization positing that complex characters, once lost in evolution, are extremely unlikely to reappear and thus the loss of complex characters is virtually always irreversible.

domain A relatively small protein segment or module (usually 100 amino acids or less) that can fold into a specific three-dimensional structure independently of other domains.

dominance Of an allele, the extent to which it produces when heterozygous the same phenotype as when homozygous; may be contrasted with a **recessive** allele, one that is phenotypically detectable only when homozygous. Dominance of a species describes the extent to which it is numerically or otherwise predominant in a community.

driven trend Also called **active trend**. A prolonged shift in the mean of a character among the species in a clade, owing to more frequent changes within species in one direction than the other. In a **passive trend**, changes in both directions would be equally likely, but are constrained by a boundary in one direction.

duplication The production of another copy of a locus (or other sequence) that is inherited as an addition to the genome.

E

ecological biogeography *See* **historical biogeography**.

ecological niche The range of combinations of all relevant environmental variables under which a species or population can persist; often more loosely used to describe the "role" of a species, or the resources it utilizes.

ecological release The expansion of a population's niche (e.g., range of habitats or resources used) where competition with other species is alleviated.

ecological speciation Speciation caused by divergent selection, by ecological factors, on characteristics that contribute to reproductive isolation.

ecotype A genetically determined phenotype of a species that is found as a local variant associated with certain ecological conditions.

effective population size The effective size of a real population is equal to the number of individuals in an ideal population (i.e., a population in which all individuals reproduce equally) that produces the rate of genetic drift seen in the real population.

electrophoresis A method of separating genetically different forms of a protein, once an important way to detect variation in the encoding genes.

endemic Of a taxon, restricted to a specified region or locality.

endosymbiont An organism that resides within the cells of a host species.

enhancer A DNA sequence that, when acted on by **transcription factors** controls transcription of an associated gene. *Cf.* ***cis*-regulatory element**, **control region**, **promoter**.

environment Usually, the complex of external physical, chemical, and biotic factors that may affect a population, an organism, or the expression of an organism's genes; more generally, anything external to the object of interest (e.g., a gene, an organism, a population) that may influence its function or activity. Thus, other genes within an organism may be part of a gene's environment, or other individuals in a population may be part of an organism's environment.

environmental correlation (r_E) *See* **genetic correlation**.

environmental sex determination The condition in which an individual's sex is determined by the environmental conditions it experiences during development, rather than (for example) its genotype. *See also* **genetic sex determination**.

environmental variance Variation among individuals in a phenotypic trait that is caused by variation in the environment rather than by genetic differences.

epigenetic inheritance Inherited changes in gene expression or phenotype that are not based on changes in DNA sequence.

epistasis An effect of the interaction between two or more gene loci on the phenotype or fitness whereby their joint effect differs from the sum of the loci taken separately.

equilibrium An unchanging condition, as of population size or genetic composition. Also, the value (e.g., of population size, allele frequency) at which this condition occurs. An

equilibrium need not be stable. *See* **stability**, **unstable equilibrium**.

ESS *See* **evolutionarily stable strategy**.

essentialism The philosophical view that all members of a class of objects (such as a species) share certain invariant, unchanging properties that distinguish them from other classes.

euploid Of a cell or organism, possessing the normal, balanced, number of chromosomes.

eusociality Animal societies characterized by overlapping generations, cooperative care of offspring (including those of other individuals), and a division of labor between reproductive and non-reproductive groups of adults.

evolution In a broad sense, the origin of entities possessing different states of one or more characteristics and changes in the proportions of those entities over time. *Organic evolution*, or *biological evolution*, is a change over time in the proportions of individual organisms differing genetically in one or more traits. Such changes transpire by the origin and subsequent alteration of the frequencies of genotypes from generation to generation within populations, by alteration of the proportions of genetically differentiated populations within a species, or by changes in the numbers of species with different characteristics, thereby altering the frequency of one or more traits within a higher taxon.

evolutionarily stable strategy (ESS) A phenotype such that, if almost all individuals in a population have that phenotype, no alternative phenotype can invade the population or replace it.

evolutionary constraint A property of organisms that tends to retard evolution of a feature or to direct its evolution along some paths rather than others.

evolutionary developmental biology (EDB) The study of evolutionary changes in the developmental bases of phenotypic characteristics.

evolutionary radiation *See* **adaptive radiation**.

evolutionary reversal The evolution of a character from a derived state back toward a condition that resembles an earlier state.

evolutionary synthesis The reconciliation of Darwin's theory with the findings of modern genetics, which gave rise to a theory that emphasized the coaction of random mutation, selection, genetic drift, and gene flow; also called the **modern synthesis**.

evolutionary trade-off The existence of both a fitness benefit and a fitness cost of a mutation or character state, relative to another.

evolutionary trend A bias in the direction of repeated changes in a character, within one lineage or among multiple lineages, over an extended period of time.

evolvability Can refer either to a measure of additive genetic variation that enables response to selection, or to the ability of genetic and developmental processes to generate potentially adaptive variation.

exaptation The evolution of a function of a gene, tissue, or structure other than the one it was originally adapted for; can also refer to the adaptive use of a previously nonadaptive trait.

exon That part of a gene that is translated into a polypeptide (protein). *Cf.* **intron**.

exon shuffling The formation of new genes by assembly of exons from two or more preexisting genes. The classical model of exon shuffling generates new combinations of exons mediated via recombination of intervening introns; however, exon shuffling can also come about by retrotransposition of exons into pre-existing genes.

exponential growth Nonlinear increase (or decrease) of a property (e.g., body size, population size) over time, described by an exponential equation.

F

fecundity The quantity of gametes (usually eggs) produced by an individual.

female choice Differential response of females to phenotypic variation in male traits, that may result in sexual selection on males; female choice does not require or imply active cognition.

Fisher's runaway A process postulated by R. A. Fisher in which a mating preference and a sexual display become very rapidly exaggerated as the result of a genetic correlation that develops between them. *See also* **indirect selection**.

fitness The success of an entity in leaving descendants to the next generation. Most often refers to the average contribution of an allele, genotype, or phenotype; can also refer to the contribution of a specific entity (e.g., an individual). *See also* **relative fitness**.

fitness component One of several events in the life cycle of many organisms that contributes to the determination of fitness, such as survival to maturity, mating success, and fecundity.

fitness function The function that relates the phenotypic value for a trait to the average fitness of individuals with that trait value.

fixation Attainment of a frequency of 1 (i.e., 100 percent) by an allele in a population, which thereby becomes **monomorphic** for the allele.

founder effect The principle that the founders of a new population carry only a fraction of the total genetic variation in the source population.

founder event A **population bottleneck** that results when a new population is founded by a small number of individuals.

founder-flush speciation A hypothesis for speciation, in which genetic change is enhanced in populations that grow rapidly ("flush") after being founded by a few individuals.

frameshift mutation An insertion or deletion of base pairs in a translated DNA sequence that alters the reading frame, resulting in multiple downstream changes in the potential gene product.

frequency In this book, usually used to mean *proportion* (e.g., the frequency of an allele is the proportion of gene copies having that allelic state).

frequency-dependent selection A mode of natural selection in which the fitness of each genotype varies as a function of its frequency in the population.

functional constraint Limitation on the variation expressed in a phenotype (perhaps a protein) because many variants have impaired function and reduce fitness.

fundamental theorem of natural selection A mathematical result derived by R. A. Fisher stating that under certain conditions, the mean fitness of a population will increase in each generation by an amount equal to the additive genetic variance for relative fitness.

G

game theory In **behavioral ecology**, a theoretical framework for analyzing the evolution of social interactions. *See* **evolutionarily stable strategy (ESS)**.

gametic selection Natural selection among alleles based on their effects in gametes.

gene The functional unit of heredity.

gene conversion A process involving the unidirectional transfer of DNA information from one gene to another. In a typical conversion event, a gene or part of a gene acquires the same sequence as the other allele at that locus (intralocus or intra-allelic conversion), or the same sequences as a different, usually paralogous, locus (interlocus conversion). One consequence of gene conversion may be the homogenization of sequences among members of a gene family.

gene copy Refers to a representative of a particular gene in an individual or cell (e.g., one copy in a haploid cell, two copies in a diploid).

gene duplication The process whereby new genes arise as copies of preexisting gene sequences. The result can be a **gene family**.

gene family Two or more loci with similar nucleotide sequences that have been derived from a common ancestral sequence.

gene flow The incorporation of genes into the gene pool of one population from one or more other populations.

gene frequency *See* **allele frequency**.

gene pool The totality of the genes of a given sexual population.

gene swamping The loss of a locally advantageous allele cause by the influx of other alleles from other populations.

gene trafficking The movement of a locus between locations in the genome that results when a gene is duplicated and the original copy of the gene is deleted or becomes a pseudogene.

gene tree A diagram representing the history by which gene copies have been derived from ancestral gene copies in previous generations.

genetic assimilation A process whereby a phenotype whose development is triggered by an environmental stimulus evolves to be constitutively expressed (i.e., no longer requires the stimulus).

genetic conflict Antagonistic fitness relationships between alleles, either at the same locus (intralocus conflict) or at different loci (interlocus conflict).

genetic constraint A restriction that prevents a lineage from evolving along a particular evolutionary trajectory because genetic variation enabling that trajectory is not available.

genetic correlation Correlated differences among genotypes in two or more phenotypic characters, due to **pleiotropy** or **linkage disequilibrium**. Genetic correlation, together with character correlation caused by different environmental conditions (**environmental correlation**), accounts for the correlation that may be observed between phenotypic characters within a population (**phenotypic correlation**).

genetic covariance The component of the phenotypic covariance between two quantitative traits that results from genetic causes; genetic covariances result from pleiotropy and linkage disequilibrium. *See also* **genetic correlation**.

genetic distance Any of several measures of the degree of genetic difference between populations, based on differences in allele frequencies.

genetic drift Random changes in the frequencies of two or more alleles or genotypes within a population.

genetic line of least resistance The combination of quantitative traits for which additive genetic variance is maximized and so will show the maximal response to directional selection.

genetic load Any reduction of the mean fitness of a population resulting from the existence of genotypes with a fitness lower than that of the most fit genotype.

genetic marker A readily detected genetic variant (such as a visible mutation or a polymorphic nucleotide) that is used to trace variation and inheritance of a closely linked region that may include a gene of interest.

genetic sex determination The condition in which an individual's sex is determined by its genotype, rather than (for example) environmental conditions. *See also* **environmental sex determination**.

genetic toolkit The set of genes and proteins, often conserved across distantly related organisms, and the developmental pathways that they comprise, by which multicellular organisms are constructed during development.

genetic variance Variation in a trait within a population, as measured by the variance that is due to genetic differences among individuals.

genic selection A form of selection in which the single gene is the unit of selection, such that the outcome is determined by fitness values assigned to different alleles. *See* **individual selection**, **kin selection**, **natural selection**.

genome The entire complement of DNA sequences in a cell or organism. A distinction may be made between the nuclear genome and organelle genomes, such as those of mitochondria and plastids.

genotype × environment interaction Phenotypic variation arising from the difference in the effect of the environment on the expression of different genotypes.

genotype frequency The proportion of individuals in a population that carry a specific genotype at one or more loci.

genotype The set of genes possessed by an individual organism; often, its genetic composition at a specific locus or set of loci singled out for discussion.

geographic variation Variation in a characteristic or allele frequency among spatially distributed populations of a species.

Gondwana The southern of the two large continents that existed in the early Mesozoic.

good genes A term used for a mechanism that may cause the evolution of mating preferences for extreme mating displays. When these displays are correlated with genetic variation for lifetime fitness, natural selection acting on those "good genes" can cause the mating preferences to become exaggerated by **indirect selection**.

grade A group of species that have evolved the same state in one or more characters and typically constitute a **paraphyletic** group relative to other species that have evolved further in the same direction.

gradualism The proposition that large differences in phenotypic characters have evolved through many slightly different intermediate states. *See* **phyletic gradualism**.

green beard effect The evolution of altruistic behavior through the evolution of a trait that simultaneously allows individuals to recognize and to help others with the same trait (e.g., a "green beard").

group selection The differential rate of origination or extinction of whole populations (or species, if the term is used broadly) on the basis of differences among them in one or more characteristics. May also refer to differences among populations in their contribution of genes to the combined gene pool. *See also* **interdemic selection**, **species selection**.

H

habitat selection The capacity of an organism (usually an animal) to choose a habitat in which to perform its activities. Habitat selection is not a form of natural selection.

habitat tracking The tendency for the geographic range of a species to shift in accordance with changes in the location of its ecological requirements, rather than adapting to environmental changes in its former range.

Haldane's rule The generalization that when only one sex manifests sterility or inviability in hybrids between species, it is the heterogametic sex (with two different sex chromosomes) that does so.

Hamilton's rule The theoretical principle that an altruistic trait can increase if the benefit to recipients, multiplied by their relationship to the altruist, exceeds the fitness cost to the altruist.

haplodiploid sex determination A form of sex determination (found in several groups of arthropods) in which unfertilized eggs develop as males and fertilized eggs as females.

haploid Of a cell or organism, possessing a single chromosome complement, hence a single gene copy at each locus.

haplotype A DNA sequence that differs from homologous sequences at one or more base pair sites.

Hardy-Weinberg Pertaining to the genotype frequencies expected at a locus under ideal equilibrium conditions in a randomly mating population.

heritability The proportion of the **variance** in a trait among individuals that is attributable to differences in genotype. Heritability in the narrow sense is the ratio of **additive genetic variance** to phenotypic variance.

hermaphroditic Performance of both female and male sexual functions by a single individual.

heterochrony An evolutionary change in phenotype caused by an alteration of timing of developmental events.

heterokaryotype A genome or individual that is heterozygous for a chromosomal rearrangement such as an inversion. *Cf.* **homokaryotype**.

heterotopy Expression of a gene or character in a different location on the body of a descendant than in its ancestor.

heterozygosity In a population, the proportion of loci at which a randomly chosen individual is heterozygous, on average. Applied to a single locus, it refers to the proportion of heterozygotes in a population. In both senses, Hardy-Weinberg equilibrium is often assumed.

heterozygote An individual organism that possesses different alleles at a locus.

heterozygous advantage The manifestation of higher fitness by heterozygotes than by homozygotes at a specific locus.

higher taxon A taxon above the species level, such as a named genus or phylum.

historical biogeography The study of historical changes in the geographic distribution of organisms, including those that affect their present distribution; **ecological biogeography** addresses current factors that affect present distributions.

historical contingency Of a dynamic system (such as a locus evolving under selection), the situation in which the course and outcome of change depend in part on initial conditions.

hitchhiking Change in the frequency of an allele due to linkage with a selected allele at another locus.

homeobox genes A large family of eukaryotic genes that contain a DNA sequence known as the homeobox. The homeobox sequence encodes a protein domain about 60 amino acids in length that binds DNA. Most homeobox genes are transcriptional regulators. *Cf.* **domain**; **Hox genes**.

homeostasis Maintenance of an equilibrium state by some self-regulating capacity of an individual.

homeotic mutation A mutation that causes a transformation of one structure into another of the organism's structures.

homeotic selector genes Genes whose expression is required for the development of an entire organ, segment, or compartment of an organism.

homokaryotype A genome or individual that is homozygous for a chromosomal rearrangement such as an inversion. *Cf.* **heterokaryotype**.

homologous chromosome *See* **homology**.

homology Possession by two or more species of a character state derived, with or without modification, from their common ancestor. **Homologous chromosomes** are those members of a chromosome complement that bear the same genes.

homonymous Pertaining to biological structures that occur repeatedly within one segment of the organism, such as teeth or bristles.

homoplasy Possession by two or more species of a similar or identical character state that has not been derived by both species from their common ancestor; embraces **convergence**, **parallel evolution**, and **evolutionary reversal**.

homozygote An individual organism that has the same allele at all of its copies of a genetic locus.

horizontal transmission Movement of genes or symbionts (such as parasites) between individual organisms other than by transmission from parents to their offspring (which is **vertical transmission**). Horizontal transmission of genes is also called **lateral gene transfer**.

Hox genes A subfamily of **homeobox genes**, conserved in all metazoan animals, that controls anterior-posterior segment identity by regulating the transcription of many genes during development.

hybrid An individual formed by mating between unlike forms, usually genetically differentiated populations or species.

hybrid zone A region in which genetically distinct populations come into contact and produce at least some offspring of mixed ancestry.

hybridization Production of offspring by interbreeding between members of genetically distinct populations.

hypermorphosis An evolutionary increase in the duration of ontogenetic development, resulting in features that are exaggerated compared to those of the ancestor.

hypothesis An informed conjecture or proposition of what might be true.

hypothetico-deductive method A scientific method in which a hypothesis is tested by deducing expected data or observations from it, if it were true, and comparing the deduced predictions with real data.

I

identical by descent Of two or more gene copies, being derived from a single gene copy in a specified common ancestor of the organisms that carry the copies.

inbreeding Mating between relatives that occurs more frequently than if mates were chosen at random from a population.

inbreeding coefficient The probability that a random pair of gene copies, inherited by offspring from two parents, is identical by descent.

inbreeding depression Reduction, in inbred individuals, of the mean value of a character (usually one correlated with fitness) relative to offspring of unrelated individuals.

inbreeding load The decline in a population's mean fitness that results from the fixation of deleterious mutations by drift. *See also* **inbreeding depression**.

inclusive fitness The fitness of a gene or genotype as measured by its effect on the survival or reproduction of both the organism bearing it (**direct fitness**) and the genes, identical by descent, borne by the organism's relatives (**indirect fitness**).

incomplete lineage sorting Persistence of a genetic polymorphism through a speciation event, so that fixation occurs only in the descendant species, or in their descendants after subsequent speciation.

indirect development A life history that includes a larval stage between embryo and adult stages. *Cf.* **direct development**.

indirect fitness *See* **inclusive fitness**.

indirect response to selection The component of evolutionary change in the mean of a trait resulting from selection that acts on other traits that are genetically correlated with it. *See also* **direct response to selection**.

indirect selection The evolution of an allele (or trait) caused by selection that acts on another locus (or trait) with which it is genetically correlated.

individual selection A form of natural selection consisting of nonrandom differences in **fitness** among different phenotypes (or genotypes) within a population. *See also* **genic selection**, **natural selection**.

individualization The evolution of distinct form and identity of each of several structures that were not differentiated from one another in an ancestor.

infanticide The killing of immature individuals by mature conspecific individuals.

ingroup *See* **outgroup**.

inheritance of acquired characteristics The formerly widespread belief that modifications of an individual during its lifetime, due to its behavior or its environment, could be transmitted to its descendants.

intelligent design (ID) A strain in creationism that claims that the complexity of organisms is too great to have evolved by natural processes and therefore must have been designed by an intelligent being.

inter-, intra- Prefixes meaning, respectively, "between" and "within." For example, "interspecific" differences are differences between species and "intraspecific" differences are differences among individuals within a species.

interaction Strictly, the dependence of an outcome on a combination of causal factors, such that the outcome is not

predictable from the average effects of the factors taken separately. More loosely, an interplay between entities that affects one or more of them (as in interactions between species). *See also* **genotype × environment interaction**.

interdemic selection **Group selection** of populations within a species.

intragenic recombination Recombination within a gene.

intrinsic rate of natural increase The potential per capita rate of increase of a population with a stable age distribution whose growth is not depressed by the negative effects of density.

introgression Movement of genes from one species or population into another by hybridization and backcrossing; carries the implication that some genes in a genome undergo such movement, but others do not.

intron A part of a gene that is not translated into a polypeptide. *Cf.* **exon**.

inversion A 180° reversal of the orientation of a part of a chromosome, relative to some standard chromosome.

isolating barrier, isolating mechanism A genetically determined difference between populations that restricts or prevents gene flow between them. The term does not include spatial segregation by extrinsic geographic or topographic barriers.

isolation by distance A model of population structure in which the likelihood of mating decreases with the geographic distance between individuals, so that local mating causes geographic variation in allele frequencies.

iteroparous Pertaining to a life history in which individuals reproduce more than once. *Cf.* **semelparous**.

K

karyotype The chromosome complement of an individual.

key adaptation An adaptation that provides the basis for using a new, substantially different habitat or resource.

kin selection A form of selection whereby alleles differ in their rate of propagation by influencing the impact of their bearers on the reproductive success of individuals (kin) who carry the same alleles by common descent.

L

Lamarckism The theory that evolution is caused by inheritance of character changes acquired during the life of an individual due to its behavior or to environmental influences.

lateral gene transfer *See* **horizontal transmission**.

Laurasia The northern of the two large continents that existed in the early Mesozoic.

lek An aggregation of males who engage in competitive mating displays; females mate at the lek but gain no direct benefits from their mates.

lethal allele An allele (usually recessive) that causes virtually complete mortality, usually early in development.

levels of selection The several kinds of reproducing biological entities (e.g., genes, organisms, species) that can vary in fitness, resulting in potential selection among them.

life history Usually refers to the set of traits that affect changes in numbers of individuals over generations, including age-specific values of survival, female reproduction, and male reproduction.

lineage A series of ancestral and descendant populations through time; usually refers to a single evolving species, but may include several species descended from a common ancestor.

lineage sorting The process by which each of several descendant species, carrying several gene lineages inherited from a common ancestral species, acquires a single gene lineage; hence, the derivation of a monophyletic gene tree, in each species, from the paraphyletic gene tree inherited from their common ancestor.

lineage-through-time plot A graph of the apparent change in number of lineages in a clade, often based on a time-calibrated phylogeny.

linkage Occurrence of two loci on the same chromosome: the loci are functionally linked only if they are so close together that they do not segregate independently in meiosis.

linkage disequilibrium The association of two alleles at two or more loci more frequently (or less frequently) than predicted by their individual frequencies.

linkage equilibrium The association of two alleles at two or more loci at the frequency predicted by their individual frequencies.

local adaptation Of an allele, trait, or population, the state of being differentially adapted to conditions that prevail in a spatially restricted area.

locus (plural: loci) A site on a chromosome occupied by a specific gene; more loosely, the gene itself, in all its allelic states.

logistic equation An equation describing the idealized growth of a population subject to a density-dependent limiting factor. As density increases, the rate of growth gradually declines until population growth stops.

M

macroevolution A vague term, usually meaning the evolution of substantial phenotypic changes, usually great enough to place the changed lineage and its descendants in a distinct genus or higher taxon. *Cf.* **microevolution**.

major transition One of several events in the history of life on Earth in which a qualitatively higher level of organization emerged.

male combat Direct physical contest between males that may result in sexual selection.

male-male competition Sexual selection that results from males' competing in various ways directly with one another, as distinct from sexual selection by **female choice**. *See also* **male combat**, **sperm (pollen) competition**.

mass extinction A highly elevated rate of extinction of species, extending over an interval that is relatively short on a geological time scale (although still very long on a human time scale).

maternal effect A nongenetic effect of a mother on the phenotype of her offspring, stemming from factors such as cytoplasmic inheritance, transmission of symbionts from mother to offspring, or nutritional conditions.

maximum likelihood (ML) A framework for statistical inference used for tasks such as the estimation of the parameters of a model or the properties of a population (such as its mean and variance, in simple cases) from data.

maximum parsimony *See* **parsimony**.

McDonald-Kreitman (MK) test A test for selection at a locus by comparing DNA sequence variation within species with the variation among species.

mean fitness The arithmetic average fitness of all individuals in a population, usually relative to some standard.

mean Usually the arithmetic mean or average; the sum of n values, divided by n. The mean value of x, symbolized as $\bar{x}$, equals $(x_1 + x_2 + \ldots + x_n)/n$.

meiotic drive A form of **segregation distortion** that occurs during meiosis and causes an allele to have greater than 50% probability of being transmitted to a gamete.

metapopulation A set of local populations, among which there may be gene flow and patterns of extinction and recolonization.

microevolution A vague term, usually referring to slight, short-term evolutionary changes within species. *Cf.* **macroevolution**.

microsatellite A short, highly repeated, untranslated DNA sequence.

migration Used in theoretical population genetics as a synonym for gene flow among populations; in other contexts, refers to directed large-scale movements of organisms that do not necessarily result in gene flow.

migration rate (m) The fraction of individuals (or gene copies) in a population that immigrated into the population from elsewhere within the current generation.

migration variance (σm^2) The square root of the mean of the squared distances between the birth places of mothers and their offspring.

mimicry Similarity of certain characters in two or more species due to convergent evolution when there is an advantage conferred by the resemblance. Common types include *Batesian* mimicry, in which a palatable *mimic* experiences lower predation because of its resemblance to an unpalatable *model*; and *Müllerian* mimicry, in which two or more unpalatable species enjoy reduced predation due to their similarity.

modern synthesis *See* **evolutionary synthesis**.

modularity The ability of individual parts of an organism, such as segments or organs, to develop or evolve independently from one another; the ability of developmental regulatory genes and pathways to be regulated independently in different tissues and developmental stages.

molecular clock The concept of a steady rate of change in DNA sequences over time, providing a basis for dating the time of divergence of lineages if the rate of change can be estimated.

monomorphic Having one form; refers to a population in which virtually all individuals have the same genotype at a locus. *Cf.* **polymorphism**.

monophyletic Refers to a taxon, or a branch of a phylogenetic tree or gene tree, that includes all the species (or genes) that descended from a common ancestor. *Cf.* **paraphyletic**, **polyphyletic**.

mosaic evolution Evolution of different characters within a lineage or clade at different rates, hence more or less independently of one another.

Muller's ratchet The process postulated by H. J. Muller in which a population's mean fitness declines when genotypes with the fewest deleterious mutations are lost by drift and cannot be recovered without recombination.

multigene family Also called "gene family," a set of distinct loci in a genome that originated from a single locus in an ancestor by duplication and sequence divergence.

multiple-niche polymorphism Stable variation at a locus owing to superior fitness of different genotypes under different conditions of a varying environment.

multiple stable equilibria *See* **stability**.

mutation An error in the replication of a nucleotide sequence, or any other alteration of the genome that is not manifested as reciprocal recombination.

mutation load The decrease in a population's mean fitness that results from deleterious mutations, relative to a hypothetical mutation-free population.

mutational variance The increment in the genetic variance of a phenotypic character caused by new mutations in each generation.

mutualism A symbiotic relation in which each of two species benefits by their interaction.

N

natural laws Consistent natural phenomena, described by statements that certain effects will always occur if specific conditions hold.

natural selection The differential survival and/or reproduction of classes of entities that differ in one or more characteristics. To constitute natural selection, the difference in survival and/or reproduction cannot be due to chance, and it must have the potential consequence of altering the proportions of the different entities. Thus natural selection is also definable as a deterministic difference in the contribution of different classes of entities to subsequent generations. Usually the differences are inherited. The entities may be alleles, genotypes or subsets of genotypes, populations, or, in the broadest sense, species. *See also* **genic selection**, **individual selection**, **kin selection**, **group selection**.

naturalistic fallacy A frequently used name for the belief that what is "natural" is morally right or good.

N_e (the effective population size) *See* **effective population size**.

neo-Darwinism Originally, the theory of natural selection of inherited variations, that denied that acquired characteristics might be inherited; often used more broadly to mean the modern theory that natural selection, acting on randomly generated particulate genetic variation, is the major, but not the sole, cause of evolution.

neofunctionalization Divergence of duplicate genes whereby one acquires a new function. *Cf.* **subfunctionalization**.

neoteny Heterochronic evolution whereby development of some or all somatic features is retarded relative to sexual maturation, resulting in sexually mature individuals with juvenile features. *See also* **paedomorphosis**, **progenesis**.

neutral alleles Alleles that do not differ measurably in their effect on fitness.

neutral theory of molecular evolution The hypothesis that most alleles that are polymorphic within populations and that become fixed do not significantly alter fitness and evolve by genetic drift.

nonadaptive evolution Evolution by substitution of neutral alleles.

nonsynonymous substitution A base pair substitution in DNA that results in an amino acid substitution in the protein product; also called **replacement substitution**. *Cf.* **synonymous substitution**.

norm of reaction The set of phenotypic expressions of a genotype under different environmental conditions. *See also* **phenotypic plasticity**.

normal distribution A bell-shaped frequency distribution of a variable; the expected distribution if many factors with independent, small effects determine the value of a variable; the basis for many statistical formulations.

nucleotide substitution The complete replacement of one nucleotide base pair by another within a lineage over evolutionary time.

O

ontogeny The development of an individual organism, from fertilized zygote until death.

operational sex ratio The relative numbers of males and females available to mate at any given time.

operon A segment of DNA containing multiple genes whose transcription is under the control of a single promoter.

optimality theory Models of adaptive evolution that assume that characters have evolved to nearly their optimum, within limits set by specified constraints.

optimum phenotype The phenotype that maximizes fitness.

organism Usually used in this book to refer to an individual member of a species.

orthologous Refers to corresponding (homologous) members of a gene family in two or more species. *Cf.* **paralogous**.

outcrossing Mating with another genetic individual. *Cf.* **selfing**.

outgroup A taxon that diverged from a group of other taxa (the **ingroup**) before they diverged from one another.

overdominance The expression by two alleles in heterozygous condition of a phenotypic value for some character that lies outside the range of the two corresponding homozygotes.

overlapping gene A gene whose coding region overlaps with another gene, which is often transcribed in an alternate reading frame.

P

paedomorphosis Possession in the adult stage of features typical of the juvenile stage of the organism's ancestor.

Pangaea The single large "world continent" formed by coalescence of land masses in the late Paleozoic.

panmixia Random mating among members of a population.

parallel evolution (parallelism) The evolution of similar or identical features independently in related lineages, thought usually to be based on similar modifications of the same developmental pathways.

paralogous Refers to the evolutionary relationship between two different members of a gene family, within a species or in a comparison of different species. *Cf.* **orthologous**.

parapatric Of two species or populations, having contiguous but non-overlapping geographic distributions. *Cf.* **allopatric**, **sympatric**.

parapatric speciation *See* **allopatric speciation**.

paraphyletic Refers to a taxon, phylogenetic tree, or gene tree whose members are all derived from a single ancestor, but which does not include all the descendants of that ancestor. *Cf.* **monophyletic**.

parent-offspring conflict A condition in which a character state that enhances fitness of offspring reduces the fitness of a parent (or vice versa).

parental investment Parental activities or processes that enhance the survival of existing offspring but whose **costs** reduce the parent's subsequent reproductive success.

parsimony Economy in the use of means to an end (*Webster's New Collegiate Dictionary*); the principle of accounting for observations by that hypothesis requiring the fewest or simplest assumptions that lack evidence; in systematics, the principle of invoking the minimal number of evolutionary changes to infer phylogenetic relationships.

parthenogenesis Virgin birth; development from an egg to which there has been no paternal contribution of genes.

passive trend *See* **driven trend**.

PCR (polymerase chain reaction) A laboratory technique by which the number of copies of a DNA sequence is increased by replication in vitro.

peak shift Change in allele frequencies within a population from one to another local maximum of mean fitness by passage through states of lower mean fitness.

peramorphosis Evolution of a more extreme character state by prolongation of development in the descendant, compared to the ancestor.

perceptual bias (sensory bias) A difference in the ability of an organism to perceive different stimuli (e.g., low vs. high frequency sounds).

peripatric Of a population, peripheral to most of the other populations of a species.

peripatric speciation Speciation by evolution of reproductive isolation in peripatric populations as a consequence of a combination of genetic drift and natural selection.

phenetic Pertaining to phenotypic similarity, as in a phenetic classification.

phenotype The morphological, physiological, biochemical, behavioral, and other properties of an organism manifested throughout its life; or any subset of such properties, especially those affected by a particular allele or other portion of the **genotype**.

phenotypic correlation *See* **genetic correlation**.

phenotypic integration Correlation between the state of two or more functionally related characteristics, so that they are advantageously matched in most individuals.

phenotypic plasticity The capacity of an organism to develop any of several phenotypic states, depending on the environment; usually this capacity is assumed to be adaptive.

phenotypic variance The **variance** (q.v.) in a trait within a population; it may include both **genetic variance** and **environmental variance**.

phyletic gradualism A term for gradual evolutionary change in features over a long period of time.

phylogenetic niche conservatism Slow evolution of the ecological requirements of a group of organisms, resulting in long-continued dependence of related species on similar resources and environmental conditions.

phylogenetic species concept (PSC) Species conceived as groups of populations that are distinguishable from other such groups.

phylogenetic tree A diagram representing the evolutionary relationships among named groups of organisms, i.e., their history of descent from common ancestors.

phylogeny The history of descent of a group of taxa such as species from their common ancestors, including the order of branching and sometimes the absolute times of divergence.

phylogeography Description and analysis of the history and processes that govern the geographic distribution of genes within species and among closely related species, analysis that may shed light on the history of the populations.

physical constraint A restriction that prevents a lineage from evolving a trait due to the properties of biological materials.

planktonic Living in open water. *Cf.* **benthic**.

pleiotropy A phenotypic effect of a gene on more than one character.

ploidy The number of chromosome complements in an organism.

point mutation A mutation that maps to a specific gene locus; in a molecular context, usually a change of a single base pair.

polygenic character A character whose variation is based wholly or in part on allelic variation at more than a few loci.

polymerase chain reaction *See* **PCR**.

polymorphic equilibrium Of allele frequencies, a stable equilibrium at which more than one allele is maintained by selection.

polymorphism The existence within a population of two or more genotypes, the rarest of which exceeds some arbitrarily low frequency (say, 1 percent); more rarely, the existence of phenotypic variation within a population, whether or not genetically based. *Cf.* **monomorphic**.

polyphenism The capacity of a species or genotype to develop two or more forms, with the specific form depending on specific environmental conditions or cues, such as temperature or day length. A polyphenism is distinct from a **polymorphism** in that the former is the property of a single genotype, whereas the latter refers to multiple forms encoded by two or more different genotypes.

polyphyletic Refers to a taxon, phylogenetic tree, or gene tree composed of members derived by evolution from ancestors in more than one ancestral taxon; hence, composed of members that do not share a unique common ancestor. *Cf.* **monophyletic**.

polyploid Of a cell or organism, possessing more than two chromosome complements.

population A group of conspecific organisms that occupy a more or less well defined geographic region and exhibit reproductive continuity from generation to generation; ecological and reproductive interactions are more frequent among these individuals than with members of other populations of the same species.

population bottleneck (bottleneck) A severe, temporary reduction in population size.

positive selection Selection for an allele that increases fitness. *Cf.* **purifying selection**.

postzygotic Occurring after union of the nuclei of uniting gametes; usually refers to inviability or sterility that confers reproductive isolation.

preadaptation Possession of the necessary properties to permit a shift to a new niche, habitat, or function. A structure is preadapted for a new function if it can assume that function without evolutionary modification.

prezygotic Occurring before union of nuclei of uniting gametes; usually refers to events in the reproductive process that cause reproductive isolation, including those that occur before mating .

primary sexual trait The gonads and closely associated structures that distinguish males and females. *See* **secondary sexual trait**.

primordium A group of embryonic or larval cells destined to give rise to a particular adult structure.

processed pseudogene A **pseudogene** that has arisen via the retrotransposition of mRNA into cDNA.

progenesis A decrease during evolution of the duration of ontogenetic development, resulting in retention of juvenile features in the sexually mature adult. *See also* **neoteny**, **paedomorphosis**.

promoter A region of DNA that initiates transcription of a gene, by binding RNA polymerase II and certain proteins (transcription factors).

provinciality The degree to which the taxonomic composition of a biota is differentiated among major geographic regions.

pseudogene A nonfunctional member of a gene family that has been derived from a functional gene. *Cf.* **processed pseudogene**.

pull of the Recent An artifact in estimating changes in diversity in the fossil record, whereby taxa that are still alive have apparently longer durations than they would if they had been counted only from fossil data, and so will inflate the count of taxa, compared to the more remote past.

punctuated anagenesis *See* **punctuated gradualism**.

punctuated equilibria A pattern of rapid evolutionary change in the phenotype of a lineage separated by long periods of little change; also, a hypothesis intended to explain such a pattern, whereby phenotypic change transpires rapidly in small populations, in concert with the evolution of reproductive isolation.

punctuated gradualism Alternating periods of slow and more rapid gradual change in a single lineage. Also called **punctuated anagenesis**.

purifying selection Elimination of deleterious alleles from a population. *Cf.* **positive selection**.

Q

quantitative genetics Genetic analysis of continuously varying characters, often employing statistical descriptions and estimators of variation.

quantitative trait A phenotypic character that varies continuously rather than as discretely different character states.

quantitative trait locus/loci (QTL) A chromosome region containing at least one gene that contributes to variation in a quantitative trait. QTL mapping is a procedure for determining the map positions of QTL on chromosomes.

quantitative variation *See* **quantitative trait**.

R

race A poorly defined term for a set of populations occupying a particular region that differ in one or more characteristics from populations elsewhere; equivalent to subspecies. In some writings, a distinctive phenotype, whether or not allopatric from others.

radiation *See* **adaptive radiation**.

radiometric dating Estimating ages of geological materials and events by the decay of radioactive elements.

random genetic drift *See* **genetic drift**.

random walk A mathematical model of a series of random fluctuations, used to describe random genetic drift and some other biological processes.

reaction norm *See* **norm of reaction**. *See also* **genotype × environment**, **phenotypic plasticity**.

realized heritability The heritability of a trait as calculated retrospectively from the change in a population's mean phenotype, relative to the selection differential that was applied to the character in an artificial selection experiment.

recessive *See* **dominance**.

reciprocal translocation A recombinational exchange of parts of two nonhomologous chromosomes.

reciprocity Cooperation based on reciprocal aid in a succession of encounters between individuals.

recombinational speciation The origin of a new species by selection among genotypes formed by hybridization between two ancestral species.

recruitment (1) In evolutionary genetics, the evolution of a new function for a gene other than the function for which that gene was originally adapted. (2) In population biology, refers to the addition of new adult (breeding) individuals to a population via reproduction (i.e., individuals born into the population that reach reproductive age).

recurrent mutation Repeated origin of mutations of a particular kind within a species.

Red Queen hypothesis The proposition that taxa become extinct at an approximately constant rate because they fail to evolve as fast as other taxa with which they have antagonistic interactions. "Red Queen" more generally refers to averting extinction by evolving as fast as possible.

refugia Locations in which species have persisted while becoming extinct elsewhere.

regression In geology, withdrawal of sea from land, accompanying lowering of sea level; in statistics, a function that best predicts a dependent from an independent variable.

regulatory modularity The property of gene regulation that allows gene expression or protein function to vary in different cells, tissues, or developmental stages of the same organism, without affecting the entire morphology or life history of the organism.

reinforcement Evolution of enhanced reproductive isolation between populations due to natural selection for greater isolation.

relatedness In **behavioral ecology**, the probability that a given individual carries the same allele as a focal individual at a given locus.

relative fitness The fitness of a genotype relative to (as a proportion of) the fitness of a reference genotype, which is often set at one; the fitness values before such standardization are **absolute fitness** values.

relict A species that has been "left behind"; for example, the last survivor of an otherwise extinct group. Sometimes, a species or population left in a locality after extinction throughout most of the region.

replacement substitution *See* **nonsynonymous substitution**.

reporter construct A DNA segment in which a putative *cis*-regulatory sequence is spliced upstream of a gene whose expression can be easily assayed, such as β-galactosidase or green fluorescent protein.

reproductive assurance Mechanisms that increase the probability of successful reproduction when potential mates are rare.

reproductive effort The proportion of energy or materials that an organism allocates to reproduction rather than to growth and maintenance.

reproductive isolation Reduction of gene exchange between populations by any of several possible factors, usually those arising from biological differences between the populations.

reproductive success The fitness of a genotype or other biological entity, often measured by the average per capita number of offspring that a newly formed zygote will have, or by similar measures.

response to selection The change in the mean value of a character over one or more generations due to selection.

restriction enzyme An enzyme that cuts double-stranded DNA at specific short nucleotide sequences. Genetic variation within a population results in variation in DNA sequence lengths after treatment with a restriction enzyme, or **restriction fragment length polymorphism** (**RFLP**).

reticulate evolution Union of different lineages of a clade by hybridization.

retrotransposition The insertion into a chromosome of a DNA sequence that originated by the reverse transcription of an RNA precursor. Used by several types of **transposons** to replicate themselves, retrotransposition is also a mechanism for gene **duplication**. *See also* **reverse transcriptase**.

reverse transcriptase An enzyme in retroviruses that synthesizes DNA copies of RNA molecules.

RFLP *See* **restriction enzyme**.

rtPCR (reverse transcriptase PCR, real-time PCR) A PCR reaction using mRNA as a template in which an initial step converts the mRNA to cDNA using reverse transcriptase and a subsequent step uses PCR to amplify the cDNA.

ruby-in-the-rubbish effect The loss of a beneficial mutation (the "ruby") that occurs in a genotype that has low fitness because of deleterious mutations (the "rubbish") that it carries at other loci.

runaway sexual selection A model of sexual selection in which a male display character and female preference for the character reinforce one another so that both evolve to be more extreme.

S

saltation A jump; a discontinuous mutational change in one or more phenotypic traits, usually of considerable magnitude.

sampling error The amount of inaccuracy (i.e., random variation) in the estimate of some value of a population, caused by measuring only a portion of the population; by extension, the chance variation in the value of repeated samples from the population.

scala naturae The "scale of nature," or Great Chain of Being: the pre-evolutionary concept that all living things were created in an orderly series of forms, from lower to higher.

scientific theory A coherent body of statements, based on reasoning and (usually) evidence, that explains some aspect of nature by recourse to natural laws or processes.

secondary contact Contact and potential interbreeding between formerly allopatric populations, owing to range expansion.

secondary sexual traits Traits not directly associated with the gonads and genitalia (the **primary sexual traits**) that differ between the sexes.

segregation distortion Any of several biological processes that alter the rules of Mendelian inheritance such that some alleles when heterozygous have a greater than 50 percent chance of transmission to the offspring. *See* **meiotic drive**.

selection Nonrandom differential survival or reproduction of classes of phenotypically different entities. *See* **natural selection**, **artificial selection**.

selection coefficient The difference between the mean relative fitness of individuals of a given genotype and that of a reference genotype.

selection differential The difference between the mean character value in a population before selection, and in the subset of individuals that survive and reproduce.

selection gradient The slope of the relationship between phenotype and fitness, for a quantitative character, usually taking correlations with other characters into account.

selection plateau The mean character value at which a population ceases to respond to continuing directional selection.

selective advantage The increment in fitness (survival and/or reproduction) provided by an allele or a character state.

selective (or functional) constraint A restriction that prevents a lineage from evolving a particular trait because that trait is always disadvantageous or interferes with the function of another trait.

selective interference Reduction in the spread of an advantageous allele that results from selection acting on other loci. *See also* **clonal interference**, **Muller's ratchet**, and **ruby-in-the-rubbish effect**.

selective sweep The increase in frequency and fixation of a beneficial allele; often used in reference to the associated reduction or elimination of DNA sequence variation in its vicinity on the chromosome.

self-incompatibility A condition in which gametes produced by a hermaphroditic individual are unable to unite, due to molecular interactions that prevent self-fertilization.

selfing Self-fertilization; union of female and male gametes produced by the same genetic individual. *Cf.* **outcrossing**.

selfish In **behavioral ecology**, a behavior that increases the fitness of the actor and decreases the fitness of one or more others.

"selfish DNA" A DNA sequence that has the capacity for its own replication, or replication via other self-replicating elements, but has no immediate function (or is even deleterious) for the organism in which it resides.

semelparous Pertaining to a life history in which individuals (especially females) reproduce only once. *Cf.* **iteroparous**.

semispecies One of several groups of populations that are partially but not entirely isolated from one another by biological factors (**isolating mechanisms**).

sensory bias *See* **perceptual bias**.

serial homology A relationship among repeated, often differentiated, structures of a single organism, defined by their similarity of developmental origin; for example, the several legs and other appendages of an arthropod.

sex-linked Of a gene, being carried by one of the sex chromosomes; it may be expressed phenotypically in both sexes.

sex ratio Often described as the proportion of males among offspring, either of an individual ("individual sex ratio") or a population ("population sex ratio").

sex role reversal A mating system in which females actively court males, often associated with male parental care.

sexual dimorphism The condition in which males and females are phenotypically distinct.

sexual isolation Reduction of gene exchange between populations because of preferential mating between individuals from the same population; also called **behavioral isolation**.

sexual reproduction Production of offspring whose genetic constitution is a mixture of those of two potentially genetically different gametes.

sexual selection Differential reproduction as a result of variation in the ability to obtain mates.

sexually antagonistic selection Selection that favors an allele or character state in one sex but a different allele or character state in the other sex.

sibling species Species that are difficult or impossible to distinguish by morphological characters, but may be discerned by differences in ecology, behavior, chromosomes, genetic markers, or other such features.

silent substitution *See* **synonymous substitution**.

single nucleotide polymorphism (SNP) Variation in the identity of a nucleotide base pair at a single position in a DNA sequence, within or among populations of a species.

sister taxa Two species or higher taxa that are derived from an immediate common ancestor, and are therefore one another's closest relatives.

special creation The idea that each species was individually created by God in much its present form.

speciation Evolution of reproductive isolation within an ancestral species, resulting in two or more descendant species.

species In the sense of biological species, the members of a group of populations that interbreed or potentially interbreed with one another under natural conditions. Also, a fundamental taxonomic category to which individual specimens are assigned, which often but not always corresponds to the biological species. *See also* **biological species**, **phylogenetic species concept**.

species hitchhiking Increase in the proportion of species with a specific trait because it is correlated with another trait that enhances speciation or reduces extinction.

species selection A form of **group selection** in which species with different characteristics increase (by speciation) or decrease (by extinction) in number at different rates because of a difference in their characteristics.

sperm (pollen) competition Competition among male gametes for fertilization.

spite In **behavioral ecology**, a behavior that decreases the fitness of both the actor and the recipient(s).

stability Often used to mean constancy; more often in this book, the propensity to return to a condition (a stable equilibrium) or to one of several such conditions (**multiple stable equilibria**) after displacement from that condition.

stabilizing selection Selection that maintains the mean of a character at or near a constant intermediate value in a population.

standard deviation The square root of the **variance**.

standing genetic variation Genetic variation that is present in a population before positive or directional selection acts to change allele frequencies. Contrasts with new mutations on which selection may act.

stasis Absence of substantial evolutionary change in one or more characters for some period of evolutionary time.

stem group *See* **crown group**.

stochastic Random. *Cf.* **deterministic**.

strata Layers of sedimentary rock that were deposited at different times.

subfunctionalization Divergence of duplicate genes whereby each retains only a subset of the several functions of the ancestral gene. *Cf.* **neofunctionalization**.

subspecies A named geographic race; a set of populations of a species that share one or more distinctive features and occupy a different geographic area from other subspecies.

substitution Usually, the complete replacement of one allele for another within a population or species over evolutionary time (*cf.* **fixation**). Sometimes refers to base pair differences in comparisons of homologous DNA sequences.

superspecies A group of **semispecies**.

symbiosis An intimate, usually physical, association between two or more species.

sympatric Of two species or populations, occupying the same geographic locality so that the opportunity to interbreed is presented. *Cf.* **allopatric**, **parapatric**.

sympatric speciation *See* **allopatric speciation**.

synapomorphy A derived character state that is shared by two or more taxa and is postulated to have evolved in their common ancestor.

synonymous substitution Fixation of a base pair change that does not alter the amino acid in the protein product of a gene; also called **silent substitution**. *Cf.* **nonsynonymous substitution**.

T

tandem repeat One of a group of adjacent duplicate copies of a DNA sequence.

taxon (plural: taxa) The named taxonomic unit (e.g., *Homo sapiens*, Hominidae, or Mammalia) to which individuals, or sets of species, are assigned. **Higher taxa** are those above the species level. *Cf.* **category**.

teleology The belief that natural events and objects have purposes and can be explained by their purposes.

tension zone A cline maintained by underdominant selection, even if relative fitnesses are uniform in space.

territory An area or volume of habitat defended by an organism or a group of organisms against other individuals, usually of the same species; territorial behavior is the behavior by which the territory is defended.

theistic evolution The belief that evolution occurs based on natural laws that were established by a deity.

theory *See* **scientific theory**.

threshold trait A characteristic that is expressed as discrete states, although the genetic variation underlying it is polygenic.

time for speciation The amount of time required for reproductive isolation to evolve, once the process starts.

trade-off The existence of both a fitness benefit and a fitness cost of a mutation or character state, relative to another.

transcription factor A protein that interacts with a regulatory DNA sequence and affects the transcription of the associated gene.

transcriptome A specified set of mRNA transcripts, such as those found in a specific cell type, under specific conditions or in the organism as a whole.

transition A mutation that changes a nucleotide to another nucleotide in the same class (purine or pyrimidine). *Cf.* **transversion**.

translocation The transfer of a segment of a chromosome to another, nonhomologous, chromosome; the chromosome formed by the addition of such a segment.

transposable element A DNA sequence, copies of which become inserted into various sites in the genome.

transposition Movement of a copy of a transposable element to a different site in the genome.

***trans*-regulatory element** A soluble molecule, usually a transcription factor protein, that binds to a *cis*-regulatory element of a gene, and is encoded by a gene located elsewhere in the genome.

transversion A mutation that changes a nucleotide to another nucleotide in the opposite class (purine or pyrimidine). *Cf.* **transition**.

trend *See* **evolutionary trend**.

two-fold cost of males The loss of fitness incurred by a sexually-reproducing genotype or population, relative to an asexual genotype, caused by producing male offspring, that themselves do not make descendants.

U

ultraconserved elements Regions of the genome that are highly conserved, sometimes at the level of 100% identity, between distantly related species. Many ultraconserved elements occur in exons that encode proteins, but others occur outside of genes and presumably have a regulatory function.

underdominance Lower fitness of a heterozygote than of both of the homozygotes for the same alleles.

unequal crossing over Recombination between nonhomologous sites on two homologous chromosomes.

uniformitarianism The proposition that natural processes that operated in the past are the same as in the present. (The term has usually implied gradual rather than catastrophic change.)

unstable equilibrium An equilibrium to which a system does not return if disturbed.

V

variability Properly, the ability of a system to vary. Often used to mean "variation."

variance (s^2, s^2, V, σ^2) The average squared deviation of an observation from the arithmetic mean; hence, a measure of variation.

vegetative propagation Production of offspring from somatic tissues, e.g., by buds.

vertical transmission *See* **horizontal transmission**.

vestigial Occurring in a rudimentary condition as a result of evolutionary reduction from a more elaborated, functional character state in an ancestor.

viability Capacity for survival; often refers to the fraction of individuals surviving to a given age, and is contrasted with inviability due to deleterious genes.

vicariance Separation of a continuously distributed ancestral population or species into separate populations because of the development of a geographic or ecological barrier.

virulence Usually, the damage inflicted on a host by a pathogen or parasite; sometimes, the capacity of a pathogen or parasite to infect and develop in a host.

W

whole-genome duplication The origin of a polyploid descendant by either the duplication of the genome of one species (**autopolyploidy**) or by hybridization between two unreduced gametes from different species (**allopolyploidy**).

wild-type The allele, genotype, or phenotype that is most prevalent (if there is one) in wild populations; with reference to the wild-type allele, other alleles are often termed mutations.

Z

zygote A single-celled individual formed by the union of gametes. Occasionally used more loosely to refer to an offspring produced by sexual reproduction.